

Library Management System

Code

```
class Library:
    def __init__(self):
        self.books = []
        self.patrons = []
        self.borrowed = []

    def add_book(self, book):
        self.books.append(book)
        print(book, "added successfully.")

    def register_patron(self, patron):
        self.patrons.append(patron)
        print(patron, "registered successfully.")

    def borrow_book(self, name, book):
        if name in self.patrons and book in self.books:
            self.books.remove(book)
            self.borrowed.append([name, book])
            print(name, "borrowed", book)
        else:
            print("Book or Patron not found.")

    def return_book(self, name, book):
        if [name, book] in self.borrowed:
            self.borrowed.remove([name, book])
            self.books.append(book)
            print(name, "returned", book)
        else:
            print("Record not found.")

    def display_books(self):
        if len(self.books) == 0:
            print("No books available.")
        else:
            print("\nAvailable Books:")
            for book in self.books:
                print(book)

def library_demo():
    lib = Library()
    lib.add_book("Python")
    lib.add_book("Java")
    lib.add_book("C++")
    lib.register_patron("Raj")
    lib.register_patron("Padma")
    lib.display_books()
    lib.borrow_book("Raj", "Python")
    lib.borrow_book("Padma", "Python")
    lib.display_books()
    lib.return_book("Raj", "Python")
    lib.display_books()
```

Output

```
Python added successfully.
Java added successfully.
C++ added successfully.
Raj registered successfully.
Padma registered successfully.
```

```
Available Books:
```

Python
Java
C++
Raj borrowed Python
Book or Patron not found.

Available Books:
Java
C++
Raj returned Python

Available Books:
Java
C++
Python